10th grade
Learning evidence 2.1: Analysis of Decisions (Practice)

Evaluated criteria

This exam evaluates the following criteria. Each criterion is evaluated across the items below.

- C2 Interprets decision alternatives, events, consequences, and states of nature.
- C3 Builds a payoff table from a description of the problem.
- C4 Explains the maximax criterion for decision making without probabilities.
- C5 Summarizes the maximin criterion for decision making without probabilities.
- C1 Describes how a problem can be formulated for optimal decision making, computes the expected value, and interprets the result.

A student passes a criterion if it is correctly applied in the solution. The overall exam result is binary (Passed/Not passed) and is determined by the set of criteria passed.

Problems

Problem 1: A community recreation center must choose a membership model for the next year: Model A (standard monthly pass), Model B (premium pass with classes), or Model C (flex pass). Demand for memberships can be high, medium, or low, and staffing costs can be favorable or unfavorable. The demand probabilities are $0{,}30$ (high), $0{,}45$ (medium), and $0{,}25$ (low). Staffing costs are independent of demand, with probabilities $0{,}55$ (favorable) and $0{,}45$ (unfavorable). Expected membership counts depend on demand and model: under high demand the center expects 640 members with Model A, 500 with Model B, and 720 with Model C; under medium demand it expects 480 with Model A, 380 with Model B, and 540 with Model C; under low demand it expects 320 with Model A, 250 with Model B, and 380 with Model C. Membership fees are 36 for Model A, 50 for Model B, and 32 for Model C per member per year. Variable staffing cost per member is 19 when costs are favorable and 27 when costs are unfavorable. Fixed annual costs are $6{,}800$ for Model A, $9{,}200$ for Model B, and $5{,}400$ for Model C. All monetary amounts in this problem can be treated as dollars. Use the combined demand and staffing outcomes as states of nature and the stated probabilities to compute expected profits.